

Explain calculation choices and reasonableness

Step 49

Explain the methods you use and decide if answers make sense.

★ Today I am learning to explain calculation choices and reasonableness.

1. Explain the Calculation

Look at each calculation. Explain the strategy (or strategies) you could use to solve it.

a) $4,785 + 2,639$

 My strategy: _____

 Why it works: _____

b) $7,394 - 2,158$

 My strategy: _____

 Why it works: _____

c) 36×24

 My strategy: _____

 Why it works: _____

d) $4,896 \div 16$

 My strategy: _____

 Why it works: _____

Think about:

- What method did you choose?
- Why did you choose it?
- How did it help you?
- Could you use a different method?

2. Choose and Explain

For each calculation, choose a method you would use. Then explain why it is a good choice.

a) $5,327 + 1,796$

Method I would use:

- ☐ Mental math ☐ Written
☐ Rounding ☐ Front-end estimate

 Why it is a good choice: _____

b) $8,450 - 3,275$

Method I would use:

- ☐ Mental math ☐ Written
☐ Rounding ☐ Front-end estimate

 Why it is a good choice: _____

c) 28×37

Method I would use:

- ☐ Mental math ☐ Written
☐ Rounding ☐ Front-end estimate

 Why it is a good choice: _____

d) $6,732 \div 24$

Method I would use:

- ☐ Mental math ☐ Written
☐ Rounding ☐ Front-end estimate

 Why it is a good choice: _____

Remember:

There is often more than one good way to solve a problem.

Choose a method that:

- is efficient
- is accurate
- makes sense for the numbers
- you understand

3. Is the Answer Reasonable?

Decide if each answer is reasonable. Explain your thinking.

a) $6,198 + 2,345 = 8,543$

 Is it reasonable? ☐ Yes ☐ No

 Explain: _____

b) $9,000 - 4,752 = 4,348$

 Is it reasonable? ☐ Yes ☐ No

 Explain: _____

c) $27 \times 48 = 1,296$

 Is it reasonable? ☐ Yes ☐ No

 Explain: _____

d) $5,832 \div 18 = 350$

 Is it reasonable? ☐ Yes ☐ No

 Explain: _____

Think about:

- What would be a close estimate?
- Can you check using rounding?
- Does the answer make sense in the situation?

Show an estimate or check (rounding / compatible numbers / inverse operation) to support your decision for any two of the above.

1. _____

2. _____

4. Compare Two Methods

Solve each calculation using two different methods. Compare your answers and explain which method you prefer and why.

a) $4,386 + 2,957$

 Method 1: _____

 Method 2: _____

 Which method do you prefer?
 Why? _____

Answer: _____

b) $7,236 \div 18$

 Method 1: _____

 Method 2: _____

 Which method do you prefer?
 Why? _____

Answer: _____

Answer: _____

Think about:

- Which method was easier? Faster? More accurate?
- Which method helped you see the math better?
- Can you teach this method to someone else?

5. Real-World Reasoning

Solve the problems. Then explain how you know your answer is reasonable.

- a) A store sells a blender for \$59.95. It is on sale for 15% off. How much will you pay?

 My calculation: _____

 Why my answer is reasonable: _____

- b) A school has 1,248 students. They are planning teams of 24 students each. How many teams can they make? How many students will be left over?

 My calculation: _____

 Why my answer is reasonable: _____

- c) A factory makes 3,675 pencils. They pack them equally into boxes of 25 pencils. How many boxes are filled? How many pencils are left?

 My calculation: _____

 Why my answer is reasonable: _____

Think about:

- What are you being asked to find?
- What is a good estimate?
- Does your answer fit the situation?
- If not, what might you check?

6. Reflect

Circle the best choice and explain.

- a) Which statement is true?
-
- A. Bigger numbers always give bigger answers.
-
- B. Reasonable answers can be checked.
-
- C. Only one method is correct.
-
- D. Rounding is never helpful.

Explain your choice: _____

- b) What is most important when solving a problem?
-
- A. Getting the answer quickly
-
- B. Explaining your strategy clearly
-
- C. Checking if the answer makes sense
-
- D. Using large numbers

Explain your choice: _____

My goal:

 One thing I will do to improve my reasoning is: _____

My Learning Check

Circle how confident you feel.



I understand and can explain.



I mostly understand.



I need more practice.



I found this hard.

Parent Observation (optional)

Please tick the boxes that apply.

- ☐ Explains strategies clearly ☐ Makes good estimates
☐ Checks reasonableness ☐ Communicates thinking
☐ Uses more than one method ☐ Needs more practice

Notes: _____

Today I learned...

1 thing I did well: _____

1 thing I will improve: _____

